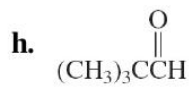
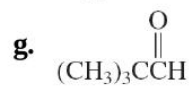
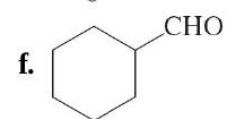
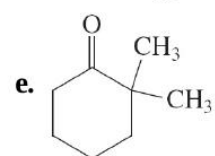
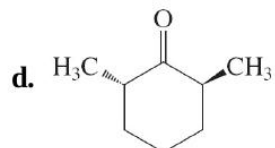
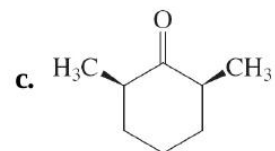
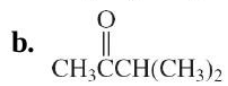
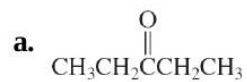


32. Underline the α -carbons and circle the α -hydrogens in each of the following structures.

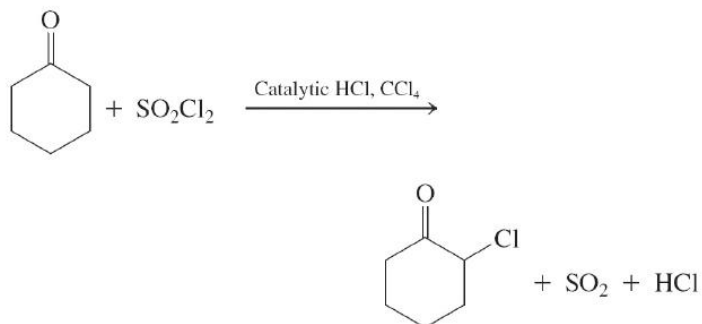


33. Write the structures of every enol and enolate ion that can arise from each of the carbonyl derivatives illustrated in [Problem 32](#).

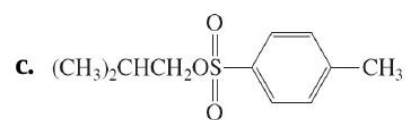
34. What product(s) would form if each carbonyl compound in [Problem 32](#) were treated with (a) alkaline D_2O ; D_2O ; (b) 1 equivalent of Br_2 in acetic acid; (c) excess Cl_2 in aqueous base?

35. Describe the experimental conditions that would be best suited for the efficient synthesis of each of the following molecules from the

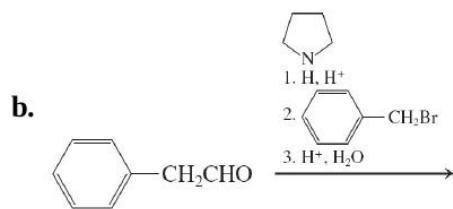
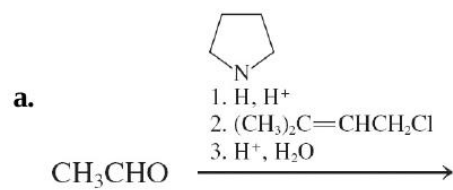
36. Propose a mechanism for the following reaction. (**Hint:** Take note of all of the products that are formed and base your answer on the mechanism for acid-catalyzed bromination of acetone shown in [Section 18-3](#).)



37. Give the product(s) that would be expected on reaction of 3-pentanone with 1 equivalent of LDA, followed by addition of 1 equivalent of



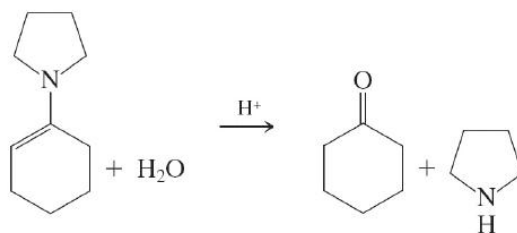
38. Give the product(s) of the following reaction sequences.



- 39.** The problem of double compared with single alkylation of ketones by iodomethane and base is mentioned in [Section 18-4](#). Write a detailed mechanism showing how some double alkylation occurs even when only one equivalent each of the iodide and base is used. Suggest a reason why the enamine alkylation procedure solves this problem.

- 40.** Would the use of an enamine instead of an enolate improve the likelihood of successful alkylation of a ketone by a secondary haloalkane?

- 41.** Formulate a mechanism for the acid-catalyzed hydrolysis of the pyrrolidine enamine of cyclohexanone (shown below).

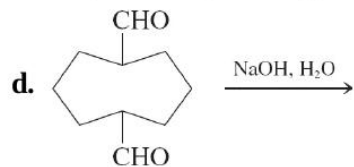
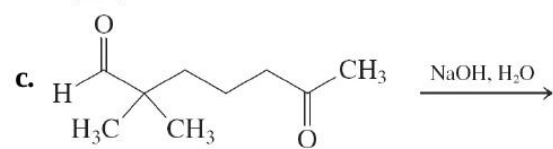
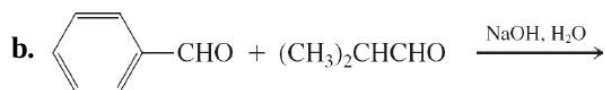
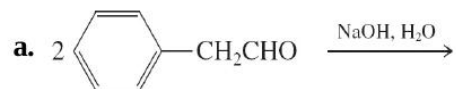


42. Write the structures of the aldol condensation products of **(a)** pentanal; **(b)** 3-methylbutanal; **(c)** cyclopentanone.

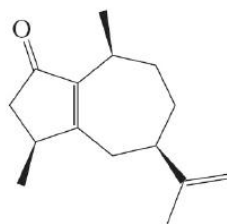
- 43.** Write the structures of the expected major products of crossed aldol condensation at elevated temperature between excess benzaldehyde and
(a) 1-phenylethanone (acetophenone—see [Section 17-1](#) for structure);
(b) acetone; **(c)** 2,2-dimethylcyclopentanone.

44. Formulate a detailed mechanism for the reaction that you wrote in [Problem 43](#), part (c).

45. Give the likely products for each of the following aldol addition reactions.



46. Rotundone (top of next column) is the natural product responsible for the peppery aroma in peppers, many herbs, and red wines ([p. 885](#)). What cyclic diketone will give rotundone upon intramolecular aldol condensation?



Rotundone

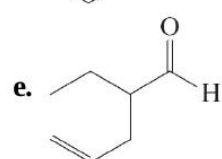
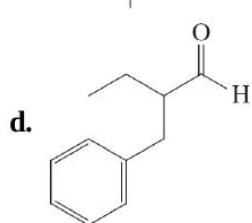
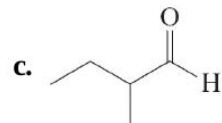
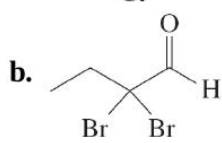
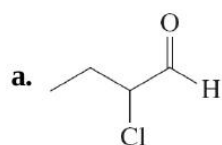
47. Write *all* possible products of the base-catalyzed crossed aldol reactions between each pair of reaction partners given below. (**Hint:** Multiple products are possible in every case; be sure to include thermodynamically unfavorable as well as favorable ones.)

a. Butanal and acetaldehyde

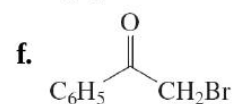
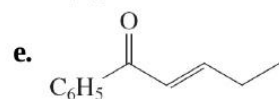
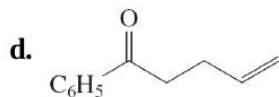
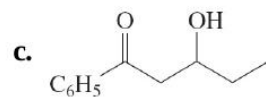
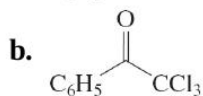
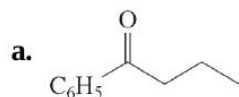
- 48.** For each of the three crossed aldol reactions described in [Problem 47](#), indicate which, if any, of the multiple possible products should predominate in the reaction mixture and explain why.

49. Aldol condensations may be catalyzed by acids. Suggest a mechanistic role for H^+ in the acid-catalyzed version. (**Hint:** Consider what kind of nucleophile might exist in acidic solution, where enolate ions are unlikely to be present.)

50. Reaction review. Without consulting the Reaction Road Maps on [pp. 852–853](#), suggest reagents to convert butanal into each of the following products.

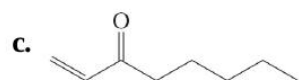
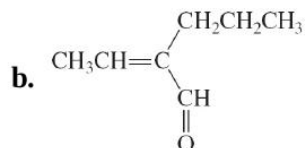
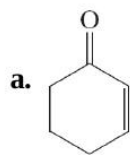


51. Reaction review II. Without consulting the Reaction Road Maps on [pp. 852–853](#), suggest reagents to convert acetophenone into each of the following substances.



52. Reaction review III. Without consulting the Reaction Road Maps on [pp. 852–853](#), suggest reagents to convert 3-buten-2-one into each of the following compounds.

55. Write the expected major product of reaction of each of the carbonyl compounds (i)–(iii) with each of the reagents (a)–(h).



a. H_2 , Pd , $\text{CH}_3\text{CH}_2\text{OH}$

b. LiAlH_4 , $(\text{CH}_3\text{CH}_2)_2\text{O}$

c. Cl_2 , CCl_4

d. KCN , H^+ , H_2O

e. CH_3Li , $(\text{CH}_3\text{CH}_2)_2\text{O}$

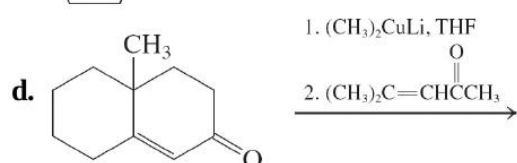
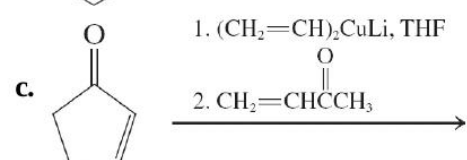
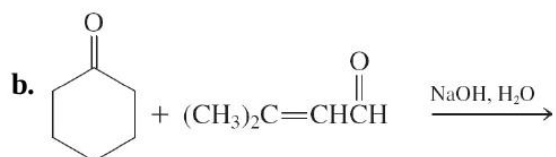
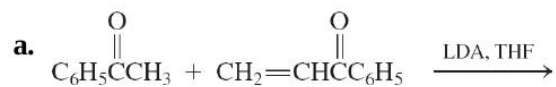
f. $(\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2)_2\text{CuLi}$, $(\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2)_2\text{CuLi}$, THF

g. $\text{NH}_2\text{NHCNH}_2$, $\text{CH}_3\text{CH}_2\text{OH}$

h. $(\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2)_2\text{CuLi}$, $(\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2)_2\text{CuLi}$, followed by treatment with $\text{CH}_2=\text{CHCH}_2\text{Cl}$ in THF

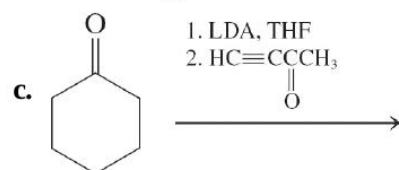
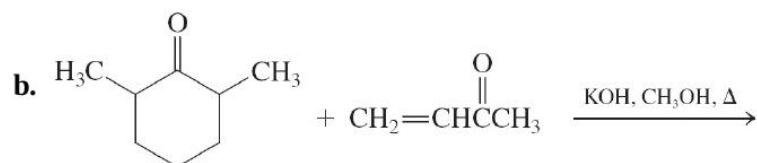
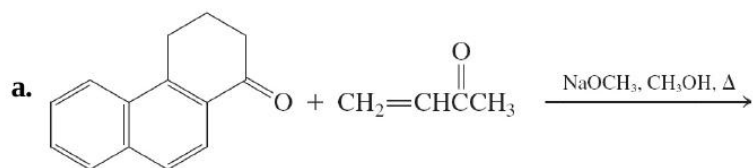
56. In each of the following retrosynthetic disconnections (~~~~), show a

57. Write the products of each of the following room temperature reactions after aqueous work-up.



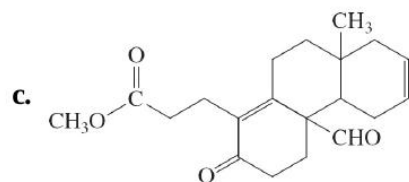
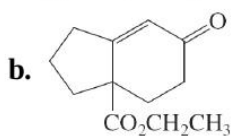
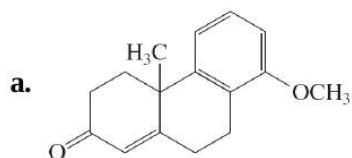
e. Write the results that you expect from base treatment of the products of reactions (c) and (d) in boiling solvent.

58. Write the final products of the following reaction sequences.



d. Write a detailed mechanism for reaction sequence (c). (**Hint:** Treat the 3-buten-2-one reagent as a Michael acceptor in the first step.)

59. Propose syntheses of the following targets by using Michael additions followed by aldol condensations (i.e., Robinson annulations). Each of the compounds shown has been instrumental in one or more total syntheses of steroidal hormones. (**Hint:** Identify and unravel the cyclohexenone ring in a reverse annulation sequence.)



60. Would you expect addition of HCl to the double bond of 3-buten-2-one (shown below) to follow Markovnikov's rule? Explain your answer by a mechanistic argument.

